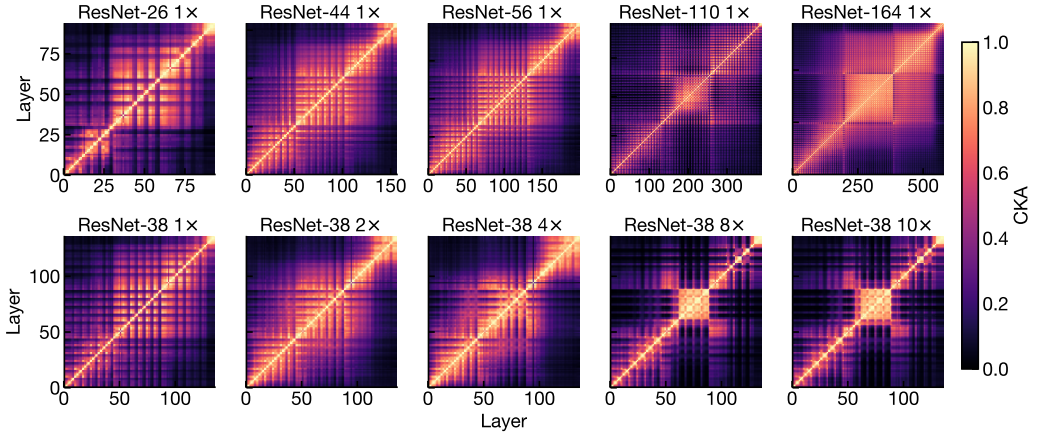

Figure D.6: How the representational structure evolves with increasing depth and width when the first principal component is removed. We plot CKA similarity heatmaps as models become deeper (top row) and wider (bottom row), with the top principal component of their internal representations removed. Compared to Figure 1, we observe that while this process significantly eliminates the block structure in large capacity models (as also shown in Figure 3), it has negligible impact on the representational structures of smaller models (where no block structure is present). The latter is not surprising, since the first principal component doesn't account for a large fraction of the variance in representations of these models, as demonstrated in Appendix Figure D.5 above.

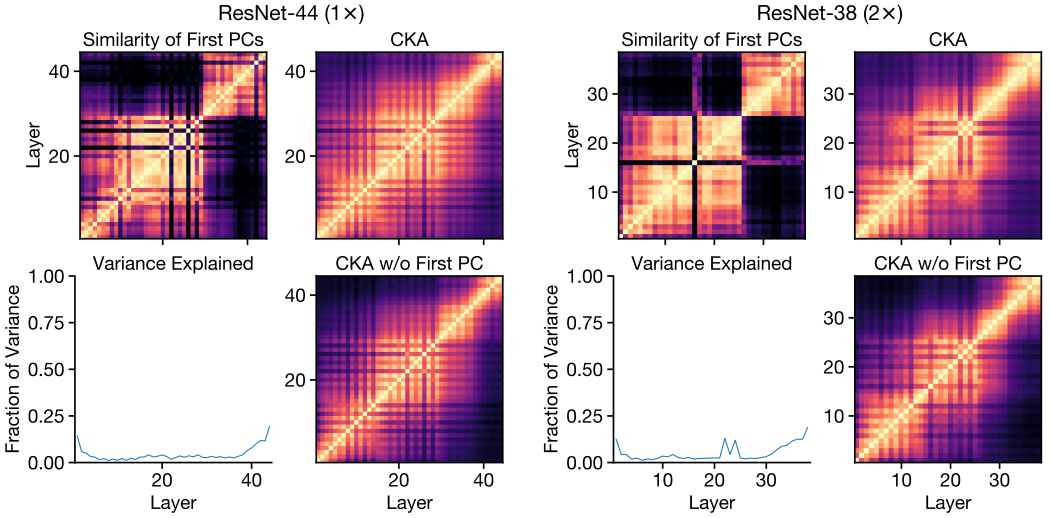

Figure D.7: Relationship between the representation similarity structure and the first principal component in networks without block structure. Above are two sets of four plots, for layers of a deep network (left) and a wide network (right), that don't contain block structure. In contrast to Figure 3, the first principal component of each layer representation only accounts for a small fraction of variance in the representation (bottom left). Comparing the squared cosine similarity of the first principal component across pairs of layers (top left), to the CKA representation similarity (top right), we find that these two structures don't resemble each other. Last but not least, the representational structure of each model remains mostly unchanged after the first principal component is removed (bottom right).

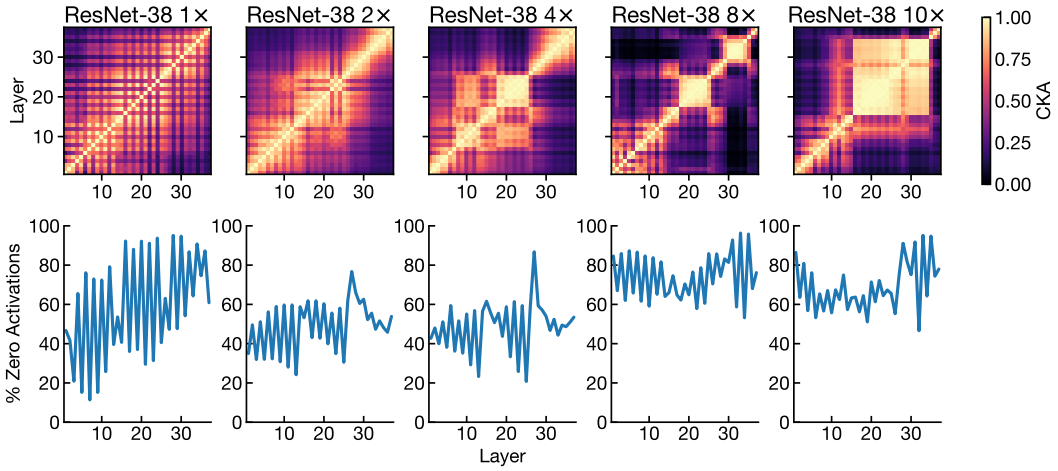

Figure D.8: ReLU activations inside and outside the block structure are similarly sparse. To rule out the possibility that the block structure arises because layers inside it behave linearly, we measured the sparsity of the ReLU activations. We observe that a significant proportion of activations are always non-zero, and the proportion is similar inside and outside the block structure. Thus, although layers inside the block structure have similar representations, each layer still applies a nonlinear transformation to its input.

E REPRESENTATIONS ACROSS MODELS

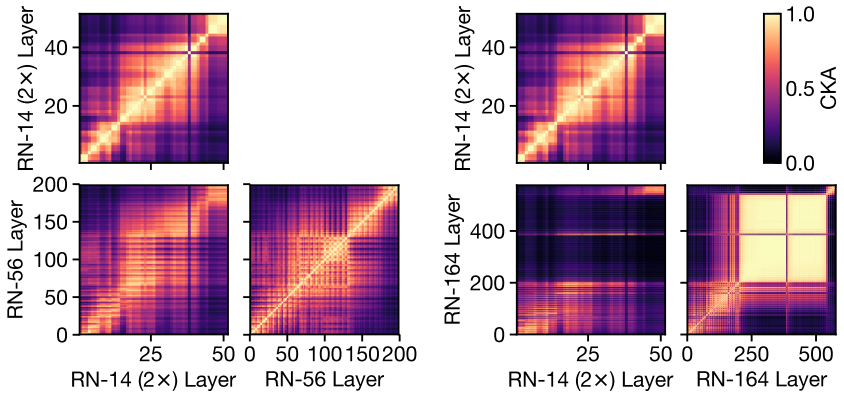

Figure E.1: Representations align between models of different widths and depths when no block structure is present. In each group of heatmaps, top left and bottom right show CKA within a single model trained on CIFAR-10. Bottom left shows CKA for all pairs of layers between these (non-architecturally-identical) models, which have similar test performance. In the absence of block structure (left group), representations at the same relative depths are similar across models. But when comparison involves models with block structure (right group), representations within the block structure are dissimilar to those of the other model.

F EXAMPLE- AND CLASS-LEVEL ACCURACY DIFFERENCES

F.1 EFFECT OF VARYING WIDTH AND DEPTH ON CIFAR-10 PREDICTIONS

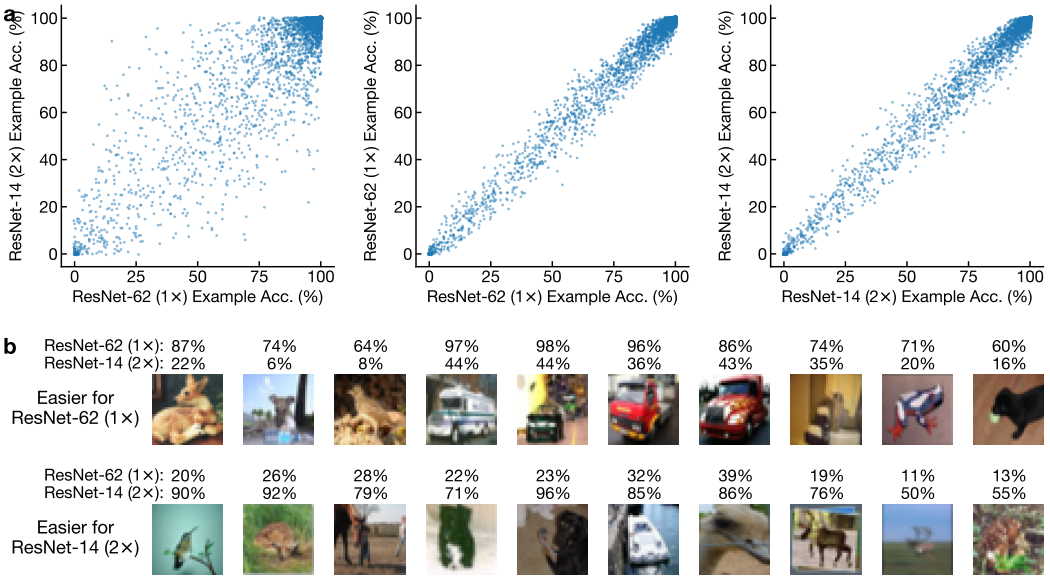

Figure F.1: Systematic per-example performance differences between wide and deep models on CIFAR-10. Comparison of predictions of 200 ResNet-62 (1×) and ResNet-14 (2×) models, which have statistically indistinguishable accuracy on the CIFAR-10 test set (mean ± SEM 94.09 ± 0.01 vs. 94.08 ± 0.01, $t(199) = 0.73$, $p = 0.47$). **a** Scatter plots of per-example accuracy for 100 ResNet-14 (2×) models vs. 100 ResNet-62 (1×) models (left) show substantially higher dispersion than corresponding plots for disjoint sets of 100 architecturally identical models trained from different initializations (middle and right). **b**: Examples with the highest accuracy differences between the two types of models. Accuracies are reported on a subset of models that is disjoint from those used to select the examples.